

EE 303: Power Systems

Course Code	EE 303
Course Name	Power Systems
L-T-P-C	3-1-0-4
Prerequisites	EE 201 Electromechanics or Instructors consent
Intended for	UG
Distribution	Compulsory for EE, Elective for CSE/ME
Approval	6th Senate (May 2014)

Course Contents

- **Module I:** Basic Concept of Three-phase circuit and Three-Phase power, Introduction to power systems and its structure: Generation, transmission and distribution, substation arrangements. Energy resources and power generation: An overview of conventional and non-conventional energy sources. [5 lectures]
- **Representation of power system components:** Generator, Transformers, Transmission lines, line parameters, transmission line design, corona, interference of power lines with communication circuits, line insulators, power cables, per unit system. [15 lectures]
- **Load Flow Analysis:** Concepts of PV, PQ and Slack Bus - NR Method [5 Lectures]
- **Fault Analysis:** Symmetrical components, symmetrical and unsymmetrical fault calculations. [8 lectures]
- **Module V:** Integrated operation of power systems, economic operation (ELD), stability, swing equation, equal area criterion, reactive power control, HVDC transmission and FACTS devices, load management. [10 lectures]
- **Module VI:** Introduction to protection and switchgear. [6 lectures]
- **Module VII:** Introduction to restructuring of power systems, power market fundamentals and price discovery, and smart grid. [4 lectures]
- **Module VIII:** Introduction to modelling and simulation to perform of power system studies/analysis through graphical user interface as well as programming based tools. Like, load flow studies, performance analysis of transmission lines, ELD, fault analysis, etc. [3 lectures]

Text Books

1. J. J. Grainger and W. D. Stevenson, **Power System Analysis**, Tata McGraw Hill.
2. Hadi Saadat, **Power System Analysis**, Tata McGraw Hill.
3. D. P. Kothari and I. J. Nagrath, **Modern Power System Analysis**, Tata McGraw Hill.
4. Ravindranath B. and Chander M., **Power System Protection and Switchgear**, New Age International Private Limited.

References

1. S. N. Singh, **Electric Power System Generation**, Transmission and Distribution, PHI.
2. NPTEL Courses
3. Paithankar Y. G. and Bhide S. R., **Fundamentals of Power System Protection**, Prentice Hall of India Private Limited.
4. O. L. Elgerd, **Electric Energy Systems Theory: An Introduction**, Tata McGraw Hill.